

GROTON-NEW LONDON, CT, USA

WMO: 725046

Lat: 41.328N

Lon: 72.049W

Elev: 3

StdP: 101.29

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 14707

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.8	-10.2	-22.5	0.5	-10.9	-20.0	0.6	-8.0	12.8	-2.5	11.7	-1.3	5.2	310	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.5	29.1	22.5	27.6	22.1	26.4	21.6	24.3	27.1	23.6	26.1	22.9	25.1	4.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.1	25.9	22.7	17.4	25.0	22.1	16.8	24.4	73.1	27.0	70.0	26.2	67.4	25.2	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.3	8.4	DB	-16.2	32.3	3.1	2.0	-18.4	33.8	-20.2	35.0	-21.9	36.1	-24.1	37.6	(4)
(5)			WB	-17.1	25.6	2.8	1.0	-19.1	26.3	-20.8	26.9	-22.3	27.5	-24.4	28.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-0.9	0.0	3.4	8.7	13.7	18.6	22.1	21.9	18.5	12.6	7.3	2.3		
	DBStd	8.93	5.33	4.53	4.25	3.45	3.24	3.05	2.49	2.53	3.21	4.18	4.25	4.61		
	HDD10.0	1260	339	281	206	64	5	0	0	0	0	22	102	241		
	HDD18.3	3096	597	514	462	290	148	33	2	3	38	182	331	496		
	CDD10.0	1531	1	0	2	24	119	259	376	369	254	103	21	3		
	CDD18.3	325	0	0	0	0	4	42	119	114	42	5	0	0		
	CDH23.3	1460	0	0	0	4	19	166	588	554	122	7	0	0		
	CDH26.7	248	0	0	0	0	2	28	118	88	11	1	0	0		
(14) Wind	WSAvg	3.9	4.5	4.4	4.6	4.3	3.6	3.2	3.0	3.0	3.6	4.2	4.3	4.4		
(15) Precipitation	PrecAvg	1204	100	92	116	107	97	87	83	102	103	94	114	114		
	PrecMax	1638	310	171	234	264	196	355	196	294	236	206	276	247		
	PrecMin	796	19	14	18	30	21	2	9	19	16	7	14	30		
	PrecStd	175	54	37	47	52	45	69	46	61	50	47	61	55		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	12.4	16.1	22.8	25.9	29.5	32.0	30.9	28.1	23.9	18.3	15.2		
		MCWB	10.9	9.0	10.4	14.1	18.7	21.8	23.8	23.8	21.6	19.5	15.1	13.5		
	2%	DB	10.3	9.8	12.9	18.6	22.8	27.2	29.4	28.9	26.1	21.9	17.0	13.2		
		MCWB	8.9	7.4	8.9	12.1	17.2	21.1	22.8	22.7	20.8	19.1	14.6	12.0		
	5%	DB	8.7	8.2	11.1	16.0	21.0	25.1	27.7	27.4	24.7	20.6	15.8	11.7		
		MCWB	7.0	6.2	8.1	10.4	16.1	19.9	22.3	22.5	20.6	18.0	13.6	10.5		
	10%	DB	7.0	6.7	9.6	14.0	19.0	23.6	26.4	26.2	23.4	19.2	14.4	10.0		
		MCWB	5.3	4.6	6.9	10.0	14.9	19.2	21.7	22.0	20.2	16.9	12.4	8.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	10.4	12.5	16.2	20.4	23.6	25.6	25.5	24.1	21.2	16.9	14.2		
		MCDB	12.0	11.1	14.0	19.5	23.9	27.6	30.1	29.0	26.5	22.6	17.8	14.6		
	2%	WB	9.2	8.3	10.5	13.9	18.6	22.4	24.3	24.4	22.9	19.9	15.6	12.2		
		MCDB	10.0	9.3	11.8	16.6	21.4	25.5	27.4	27.1	24.4	21.2	16.5	12.7		
	5%	WB	7.5	6.8	8.9	12.4	17.3	21.2	23.5	23.7	22.0	18.8	14.2	10.7		
		MCDB	8.3	7.8	10.3	14.3	19.9	23.8	26.2	26.1	23.5	20.2	15.2	11.4		
	10%	WB	5.4	5.0	7.5	11.1	16.0	20.2	22.7	23.0	21.2	17.4	12.8	8.6		
		MCDB	6.8	6.5	9.1	12.9	18.2	22.5	25.3	25.1	22.7	18.8	14.0	9.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.2	8.2	8.7	8.3	7.8	7.5	7.7	8.5	8.8	8.6	7.7		
		MCDBR	9.2	9.7	9.9	11.6	11.0	10.0	8.9	8.4	8.8	8.9	8.7	8.6		
		MCWBR	8.4	8.1	7.2	6.7	6.3	5.6	4.4	4.5	5.6	6.6	7.5	8.9		
		MCWBR	9.0	9.3	8.7	9.2	8.6	7.3	6.6	6.7	7.4	7.7	7.7	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.306	0.315	0.337	0.376	0.424	0.465	0.488	0.453	0.382	0.345	0.316	0.304		
		taud	2.496	2.472	2.454	2.375	2.270	2.196	2.143	2.233	2.406	2.519	2.554	2.552		
	Ebn at Noon		852	899	914	895	855	817	794	812	854	853	836	827		
		Edh at Noon	74	88	100	117	133	143	150	133	105	84	70	65		
(44) All-Sky Solar Radiation	RadAvg		1.86	2.74	3.75	4.87	5.42	5.91	6.03	5.40	4.42	2.98	2.05	1.56		
	RadStd		0.15	0.26	0.29	0.52	0.53	0.59	0.33	0.30	0.34	0.28	0.16	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		+0.75	N/A	N/A	+0.84	N/A	N/A	N/A	N/A	+135	N/A		
(47) Regional (76 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+102	+55		

Nomenclature: See separate page